B.TECH CSE — 5th SEMESTER LECTURE NOTES

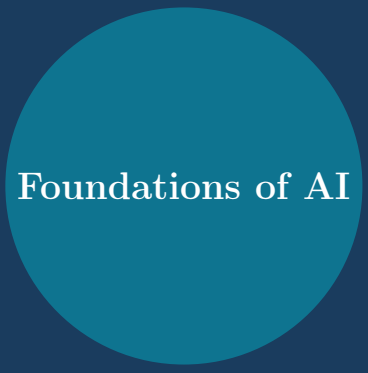
Chapter 1

Foundations of Artificial Intelligence

Course: *Fundamentals of Machine Intelligence 1 (MTH 3647)*

Based on: *Artificial Intelligence and Machine Learning*

Pradeep Singh, Tapan K. Gandhi, Balasubramanian Raman



Contents

1 Foundations of Artificial Intelligence **2**

1.1 Defining Artificial Intelligence . . . 2
1.1.1 Applications of AI . . . 2
1.1.2 What is an "Agent" ? . . . 2
1.1.3 The process of learning . . . 2
1.2 History of AI Development . . . 3
1.3 AI Concepts and Terminology . . . 3
1.3.1 Models . . . 3
1.3.2 Machine Learning (ML) . . . 3
1.3.3 Deep Learning (DL) . . . 4
1.4 Types of AI Systems . . . 4
1.4.1 Reactive Machines . . . 4
1.4.2 Limited Memory Systems . . . 4
1.4.3 Theory of Mind Systems . . . 4
1.4.4 Self Aware AI . . . 5
1.5 Strong and Weak AI . . . 5
1.5.1 Strong AI . . . 5
1.5.2 Weak AI . . . 6
1.5.3 Chinese Room Argument . . . 6
1.6 Introduction to AI Techniques . . . 6
1.6.1 Supervised Learning . . . 6
1.6.2 Unsupervised Learning . . . 6
1.6.3 Evolutionary Algorithms (EA) . . . 6
1.7 Key Components of AI systems . . . 7
1.7.1 Knowledge Representation (KR) . . . 7
1.7.2 Reasoning and Inference . . . 7
1.7.3 Learning Mechanisms . . . 7
1.7.4 Perception and Interaction . . . 8
1.8 Questions for self-study . . . 8

Foundations of Artificial Intelligence

1.1 Defining Artificial Intelligence

AI is the study of agents that perceive and act in an environment to maximize a utility function.

1.1.1 Applications of AI

- Applications of AI are vast, spanning over domains such as Computer Vision, NLP, Robotics and Automated Reasoning.
- Also seen in the development of self-driving cars, medical diagnosis etc.

1.1.2 What is an "Agent" ?

An agent(A) is defined as a system that takes input from environment using a percept and processes them using a function (f) to generate an output.

The function is a learned/optimised process which is used by the agent to interact with the environment.

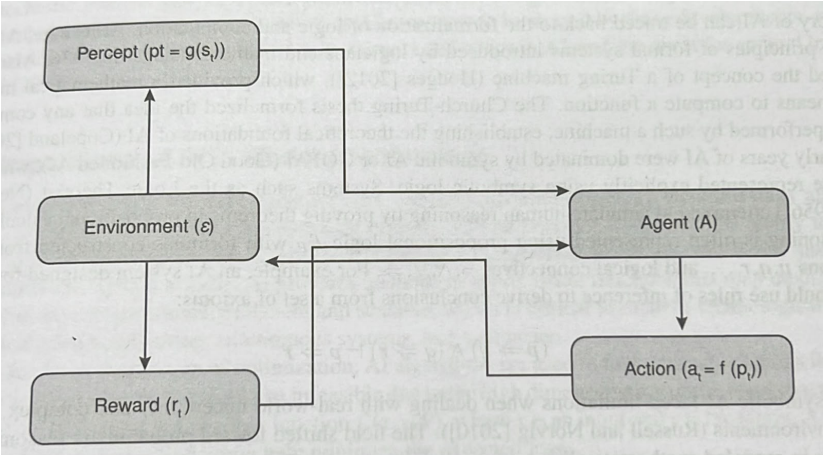


Figure 1.1: AI Framework

1.1.3 The process of learning

The process of learning can be described as approximating a function that maps from an input space(X) to an output space(Y) typically using data.

In supervised learning it involves minimising a loss function over a set of labelled data where $x_i \in X$ and $y_i \in Y$ is the corresponding label for x_i.

Mathematically represented as

$$f^* = \arg\min_f \sum_{i=1}^{N} \mathcal{L}\left(f(x_i), y_i\right)$$

1.2 History of AI Development

History of AI can be traced back to formalization of logic and computation. AI at it's core is built upon the principles of formal systems introduced by logicians and mathematicians.

Symbolic AI is the era of AI where problems were modeled/represented/expressed explicitly using symbolic logic.

It faced limitations when dealing with real-world uncertainty and complex non-deterministic environments

Uncertainty was then modeled mathematically using probability theory which was introduced in the form of Bayesian Networks.

The rise of Machine Learning shifted the focus from handcrafting knowledge representations to learning from data.

Next came deep learning which involved stacking of layers of non-linear transformations to learn complex functions.

Current state of AI is characterized by widespread application of DL models, combined with reinforcement learning and probabilistic reasoning.

1.3 AI Concepts and Terminology

1.3.1 Models

Models in AI refer to the mathematical representation of systems that map from inputs to outputs. A model $f_\theta(x)$ is parameterized by θ and attempts to approximate a true function $g_(x)$.

Training a model means to optimise θ such that expected loss $E[L(f_\theta(x), y)]$ is minimised.

1.3.2 Machine Learning (ML)

ML is a subset of AI that deals with learning patterns from data. It can also be described as a function approximation problem where we seek to find a function f_θ that

maps input x to an output label y such that the discrepancy between the predicted outputs and true labels is minimized. This is achieved using a variety of algorithms that range from linear models to complex neural networks.

1.3.3 Deep Learning (DL)

DL is a subset of ML that uses deep neural networks to model complex hierarchical structures in data.

At its core it uses multiple layers of transformations to map inputs to outputs. Each layer applies a non-linear transformation to the data and is used to capture complex patterns.

A neural network of L layers can be expressed mathematically as

$$f_\theta(x) = f_L(f_{L-1}(f_{L-2}(...f_2(f_1(x)))))$$

1.4 Types of AI Systems

1.4.1 Reactive Machines

These are primitive systems that react based solely on the current percept, without depending on past experience or memory. Examples include DFAs (Deterministic Finite Automata) and NFAs (Non-Determinisitc Finite Automata), where transitions between states depend solely on the current symbol the state receives.

1.4.2 Limited Memory Systems

These systems are formed by extending reactive systems with memory. They remember a finite set of states. Examples include Push Down Automata (PDAs) and Turing Machines.

1.4.3 Theory of Mind Systems

Machines not only understand and react to environment but also model thoughts belief of other agents in that environment.

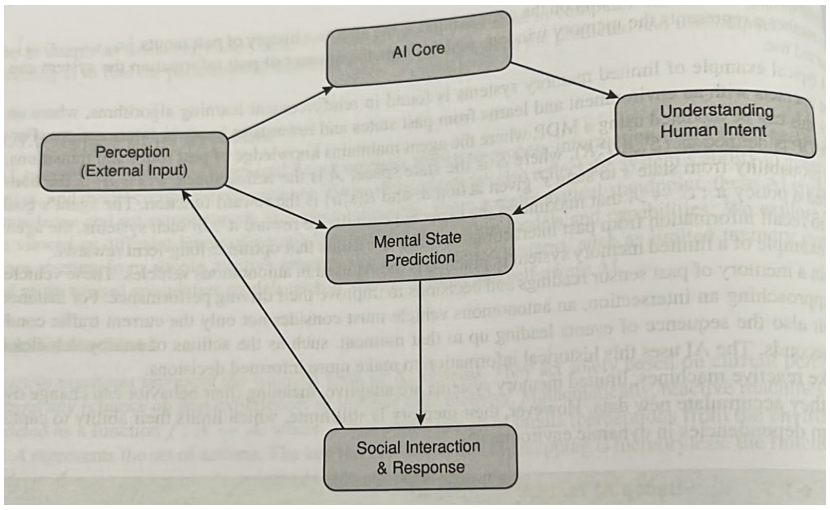


Figure 1.2: Theory of Mind AI

1.4.4 Self Aware AI

The most speculative form of AI. It does everything the Theory of Mind AI does, but also is aware of its self existence, conscience, understanding and has an emotion of its own.

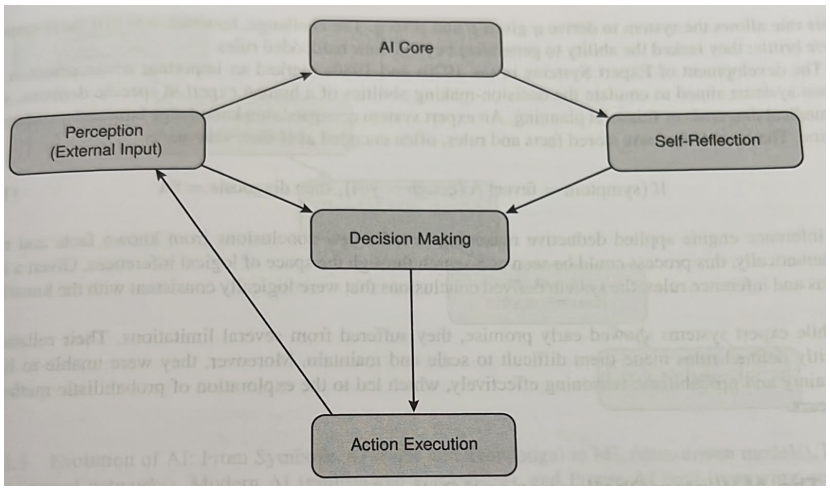


Figure 1.3: Self Aware AI

1.5 Strong and Weak AI

1.5.1 Strong AI

Strong AI also known as Artificial General Intelligence (AGI) is the concept of creating a machine that posses the abilities comparable to a human being.

Challenges in Building Strong AI

One of the primary difficulties is the need for common sense reasoning.

Another challenge is developing explainable AI systems.

Ethical Implications of Strong AI

Potential for AI systems to act autonomously in ways that may conflict with human values.

AI system's objective must align with human values remain unsolved.

Development of strong AI poses the risk of unintended consequences if system operate autonomously without proper oversight.

1.5.2 Weak AI

Weak AI refers to system designed to perform specific tasks or solve particular problems without possessing general cognitive abilities.

The primary difference between them lies in it's generalization ability.

It is widely used in supervised learning.

In computer vision Weak AI is used in the form of CNNs.

In NLP, weak AI is used in machine translation, sentiment analysis and text summarization.

1.5.3 Chinese Room Argument

A person who does not understand Chinese follows a set of instructions to manipulate Chinese symbols in response to input. From outside it may appear that the person knows Chinese. A machine can do this as well.

1.6 Introduction to AI Techniques

1.6.1 Supervised Learning

A model is trained on labelled data. The objective in supervised learning is to learn a mapping such that the predicted output closely match the true label for unseen inputs.

1.6.2 Unsupervised Learning

Deals with unlabelled data where goal is to discover hidden patterns, structures or representations without explicit supervision.

1.6.3 Evolutionary Algorithms (EA)

EAs are a class of optimisation algorithms inspired by the process of natural selection. These algorithms operate on a population of candidate solutions, evolving over time through mutation, crossover and selection.

Mathematically an evolutionary algorithm maintains a population $P = \{\theta_0, \theta_1, \theta_2, \theta_3, ..\theta_N\}$. The fitness of each solution is evaluated using a fitness function F : P -> R, and the goal is to maximize or minimize this fitness function. The algorithm then proceeds with the following steps.

1. **Selection**: Select the subset of the population based on their fitness. Solutions with higher fitness are more likely to be selected for reproduction. The selection can be formalized using probabilities.

2. **Crossover**: Combine pairs of solutions(parents) to produce new solutions(offspring). This involves exchanging genetic material between parents to create diversity in the offspring.

3. **Mutation**: Apply small random changes to the offspring to explore new areas of search space. Mutation is crucial for maintaining diversity in the population and preventing premature convergence to local optima.

4. **Replacement**: Replace the old population with the new population of offspring and repeat the process for several generations until convergence is achieved.

1.7 Key Components of AI systems

1.7.1 Knowledge Representation (KR)

KR refers to the methods and structures used to encode information about the world in a way that an AI system can utilize for reasoning and decision-making. Mathematically KR involves creating a mapping from real-world entities, relationships, events to symbolic structures that can be manipulated by algorithms. These structures often take the form of graphs, logic-based systems or probabilistic models.

1.7.2 Reasoning and Inference

RI refers to the process through which AI systems draw conclusions from information. Reasoning can be defined as the process of deriving new propositions from a set of propositions using rules of inference.

In probabilistic reasoning inference is performed by calculating posterior probabilities of unknown variables given known data often using Bayes' Theorem.

More advanced inference techniques are used in systems that need to reason over complex KBs such as constraint satisfaction problems(CSP).

1.7.3 Learning Mechanisms

Learning Mechanisms allow AI models to improve performance over time by adapting new information. These mechanisms are grounded in optimisation and statistical in-

ference with the objective of minimising a loss function that quantifies the differnce between model predictons and true outcomes.

In supervised learning the goal is to learn a function f : X -> Y that maps inputs to outputs based on labeled training data D. The learning process involves minimising a loss function.

In unsupervised learning AI systems learn from data without labelled outcomes. A common task in unsupervised leanring is clustering, where the objective is to group similar datapoints together.

1.7.4 Perception and Interaction

For an AI system to interact with the real world, it must have the ability to perceive its environment and act upon it. Perception involves processing raw sensory data and converting it into a structured form that the system can use for decison making.

Interaction involves the system's ability to perform actions based on its perceptions. In robotics this might involve using control algorithms to move actuators or manipulate objects. The system's action are typically determined by optimising a control policy that maps sensor inputs to motor commands.

1.8 Questions for self-study

1. Explain the benefits of AI in
   - Healthcare and medicine
   - Education and personalised learning
   - Smart cities and infrastructure.
2. AI systems can be biased. Explain
3. Explain how AI systems contribute to job displacement.
4. Explain how AI system introduce security and privacy concerns.
5. Describe the ethical issues in AI in
   - Decision Making
   - Transparency and explainability
   - Regulation and governance of AI systems.
   - Autonomous Weaponry